

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 24

Control Plane

صفحه کنترل



Lecture Notes

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1 Control Plane Concept

The **Control Plane** (C-Plane) handles all **signaling** in the mobile network — the messages that set up, manage, modify, and tear down connections. It does NOT carry user data.

Control Plane responsibilities:

- **Authentication & Security:** Verify subscriber identity, establish encryption
- **Registration & Mobility:** Attach, detach, tracking area updates, handovers
- **Session Management:** Establish/modify/release PDU sessions or bearers
- **QoS Negotiation:** Agree on quality parameters for data flows
- **Paging:** Locate idle UEs for incoming data/calls
- **Policy & Charging:** Apply operator rules to sessions

Key principle: The control plane “programs” the user plane — it sets up the tunnels, rules, and routing that the user plane then uses to forward data.

2 Control Plane Protocols – MAJOR Reference Table

Protocol	Generation	Layer	Plane	Between	Purpose
MAP (Mobile Application Part)	2G/3G	Application	Control	MSC↔HLR, VLR↔HLR	Subscriber data, auth, SMS routing, HO
ISUP (ISDN User Part)	2G/3G	Application	Control	MSC↔MSC, MSC↔PSTN	Circuit-switched call setup/release
RANAP (RAN Application Protocol)	3G	Application	Control	RNC↔CN (MSC/SGSN)	RAN-Core signaling (Iu interface)
S1AP (S1 Application Protocol)	4G	Application	Control	eNB↔MME	RAN-Core signaling (S1-MME)
NGAP (NG Application Protocol)	5G	Application	Control	gNB↔AMF	RAN-Core signaling (N2)
GTP-C v1	3G	Tunnel Control	Control	SGSN↔GGSN	PDP context create/modify/delete
GTP-C v2	4G/5G	Tunnel Control	Control	MME↔S-GW↔P-GW	Session create/modify/delete bearer
Diameter	4G	AAA/Policy	Control	MME↔HSS, PCRF↔P-GW	Auth (S6a), policy (Gx), charging (Gy)
HTTP/2 SBI	5G	Service-Based	Control	NF↔NF (all 5GC NFs)	Service discovery, registration, session mgmt
NAS EMM	4G	Non-Access Stratum	Control	UE↔MME	Attach, TAU, authentication, security
NAS ESM	4G	Non-Access Stratum	Control	UE↔MME (→P-GW)	Bearer setup/modify/release
NAS 5GMM	5G	Non-Access Stratum	Control	UE↔AMF	Registration, authentication, service request
NAS 5GSM	5G	Non-Access Stratum	Control	UE↔SMF (via AMF)	PDU session establish/modify/release
RRC (LTE)	4G	Access Stratum	Control	UE↔eNB	Connection setup, meas config, HO, SIBs

Protocol	Generation	Layer	Plane	Between	Purpose
RRC (NR)	5G	Access Stratum	Control	UE↔gNB	Same + BWP config, beam mgmt, CHO
SCTP	4G/5G	Transport	Control	eNB↔MME, gNB↔AMF	Reliable signaling transport (multi-stream)
SIP (Session Initiation Protocol)	4G/5G (IMS)	Application	Control	UE↔P-CSCF↔S-CSCF	VoLTE/VoNR call setup, IMS registration
PCF	5G (4G CUPS)	Control	Control	SMF↔UPF	User plane rule programming (N4)

Protocol Stack Relationships

APPLICATION PROTOCOLS		
MAP RANAP S1AP NGAP GTP-C Diameter HTTP/2 SBI		
TRANSPORT PROTOCOLS		
SCTP	TCP	UDP
IP		
L2 / L1 (Ethernet, etc.)		

Figure 24.1 How the signalling protocols stack on transport protocols

3 NAS (Non-Access Stratum)

NAS protocols operate **between the UE and the core network** — they pass transparently through the RAN (the eNB/gNB does not interpret NAS messages, only relays them).

3.1 LTE NAS (TS 24.301)

LTE NAS has two sub-layers:

EMM (EPS Mobility Management):

Procedure	Messages	Purpose
Attach	Attach Request/Accept/Complete	Register UE to network
Detach	Detach Request/Accept	Deregister UE
TAU	TAU Request/Accept	Update Tracking Area
Authentication	Auth Request/Response	Mutual authentication (AKA)
Security Mode	SMC Command/Complete	Activate NAS ciphering/integrity
Service Request	Service Request/Accept	Transition IDLE→CONNECTED
Identity	Identity Request/Response	Request IMSI if no GUTI

ESM (EPS Session Management):

Procedure	Messages	Purpose
PDN Connectivity	PDN Conn Request/Activate Default Bearer	Establish PDN connection
Bearer Resource Alloc	Bearer Resource Alloc Req	UE-initiated dedicated bearer
Activate Dedicated Bearer	Act Ded Bearer Req/Accept	Network-initiated QoS bearer
Modify Bearer	Modify Bearer Req/Accept	Change bearer QoS
Deactivate Bearer	Deact Bearer Req/Accept	Release bearer

3.2 5G NAS (TS 24.501)

5G NAS also has two sub-layers but with enhanced capabilities:

5GMM (5G Mobility Management):

Procedure	Messages	Purpose
Registration	Registration Req/Accept/Complete	Initial, mobility, periodic reg
Deregistration	Dereg Request/Accept	UE or network initiated
Authentication	Auth Request/Response/Failure	5G-AKA or EAP-AKA'
Security Mode	SMC Command/Complete	NAS security activation
Service Request	Service Request/Accept	Resume from IDLE/INACTIVE
Configuration Update	Config Update Command/Complete	GUTI reallocation, NSSAI
Notification	Notification/Response	Network triggers (SOR, UPU)

5GSM (5G Session Management):

Procedure	Messages	Purpose
PDU Session Establishment	PDU Session Est Req/Accept	Create PDU session
PDU Session Modification	PDU Session Mod Req/Accept	Modify QoS flows
PDU Session Release	PDU Session Rel Req/Accept	Tear down session

Key 5G NAS enhancements vs. 4G:

- Registration replaces Attach + TAU (unified procedure)
- Supports network slicing (requested NSSAI in registration)
- PDU session types: IPv4, IPv6, IPv4v6, Ethernet, Unstructured
- NAS messages routed via AMF to SMF for session management

3.3 NAS Transparency Through RAN

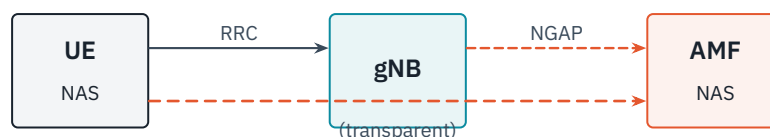


Figure 24.2 NAS passes transparently through the gNB

- NAS messages are carried inside RRC messages on the radio interface
- RRC carries them inside NGAP (5G) or S1AP (4G) to the core
- The RAN node does NOT decrypt or interpret NAS content
- Separate NAS security context (NAS ciphering/integrity) from AS security (RRC/UP)

4 RRC (Radio Resource Control)

RRC operates **between the UE and the base station** (Access Stratum) and controls the radio connection.

4.1 LTE RRC (TS 36.331)

Key Procedures:

Procedure	Messages	Purpose
Connection Setup	RRC Conn Setup Req/Setup/Complete	Establish SRB1
Connection Reconfiguration	RRC Conn Reconfig	Add/modify bearers, handover
Connection Release	RRC Conn Release	Release radio resources
Measurement Configuration	(in Reconfig)	Configure meas events (A1-A6, B1-B2)
Handover	RRC Conn Reconfig (mobilityCtrlInfo)	Execute handover
Security Mode	Security Mode Command/Complete	Activate AS ciphering/integrity
System Information	MIB, SIB1-SIB19	Broadcast cell parameters

LTE RRC States:



Figure 24.3 LTE RRC states

4.2 NR RRC (TS 38.331)

NR RRC includes all LTE RRC functions **plus**:

Feature	Description
BWP Configuration	Configure Bandwidth Parts (active BWP switching)
Beam Management	SSB beam indication, beam failure recovery, L1-RSRP reporting
Conditional Handover (CHO)	Pre-configure HO conditions — execute when met (no HO command delay)
Cell Group Config	MCG + SCG for dual connectivity
MeasConfig enhancements	NR events (similar + inter-RAT NR↔LTE), L3 filtering per beam
SIB enhancements	On-demand SIBs, SIB for RedCap, NTN, sidelink

4.3 RRC States in 5G (Three States)

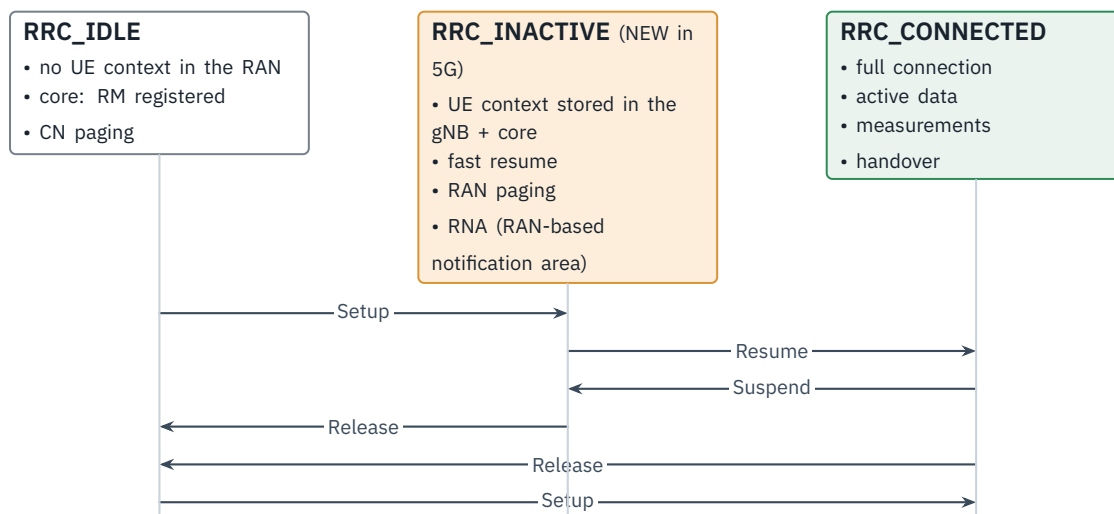


Figure 24.4 The three NR RRC states and their transitions

RRC_INACTIVE benefits:

- UE context preserved in Last Serving gNB → fast resume (~10 ms vs ~50 ms)
- No full connection setup needed — just RRC Resume
- RNA (RAN-based Notification Area) for mobility without core notification
- Ideal for IoT devices with infrequent small data

5 NGAP (NG Application Protocol) – TS 38.413

NGAP is the **5G equivalent of S1AP** (LTE) – it runs between the **gNB and AMF** over the **N2 interface**.

5.1 Transport

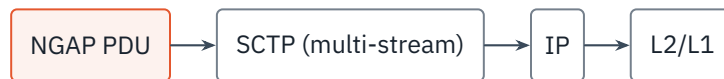


Figure 24.5 NGAP transport stack

- SCTP provides reliable, in-order delivery with multiple streams
- Stream 0: non-UE-associated signaling (setup, reset)
- Other streams: UE-associated signaling (one stream per UE-group)

5.2 Key Procedures

Category	Procedure	Description
Interface Mgmt	NG Setup	gNB registers with AMF (capabilities, served cells)
	AMF Configuration Update	AMF informs gNB of changes
	NG Reset	Error recovery
UE Context	Initial Context Setup	AMF→gNB: create UE context, security, PDU sessions
	UE Context Release	Release UE resources at gNB
	UE Context Modification	Modify existing context
Mobility	Handover Required	gNB→AMF: request inter-gNB HO
	Handover Request	AMF→target gNB: prepare HO
	Handover Notify	Target gNB confirms HO complete
	Path Switch Request	Intra-AMF, X2-based HO notification
NAS Transport	Initial UE Message	First NAS message from UE (gNB→AMF)
	Downlink NAS Transport	AMF→gNB→UE: carry NAS message
	Uplink NAS Transport	UE→gNB→AMF: carry NAS message
Paging	Paging	AMF→gNB: page UE in TA cells
PDU Session	PDU Session Resource Setup	Establish user plane resources
	PDU Session Resource Modify	Modify QoS/resources
	PDU Session Resource Release	Release UP resources

5.3 NGAP vs. S1AP Comparison

Feature	S1AP (4G)	NGAP (5G)
Interface	S1-MME	N2
Nodes	eNB ↔ MME	gNB ↔ AMF
Transport	SCTP	SCTP
Session concept	E-RAB (bearer)	PDU Session + QoS Flows
Slice support	No	Yes (S-NSSAI in procedures)
UE states	IDLE, CONNECTED	IDLE, INACTIVE, CONNECTED
Multi-connectivity	No	Yes (secondary RAT)

6 GTP-C (GTP Control Plane)

6.1 GTP-C v1 (3G – TS 29.060)

Interface	Between	Purpose
Gn/Gp	SGSN ↔ GGSN	PDP Context management

Key messages:

- Create PDP Context Request/Response
- Update PDP Context Request/Response
- Delete PDP Context Request/Response

6.2 GTP-C v2 (4G – TS 29.274)

GTPv2-C is a complete redesign with improved efficiency:

Interface	Between	Purpose
S11	MME ↔ S-GW	Session/bearer management
S5/S8	S-GW ↔ P-GW	Session/bearer management
S10	MME ↔ MME	Inter-MME handover
S3	MME ↔ SGSN	3G↔4G mobility

Key GTPv2-C Messages:

Message	Direction	Purpose
Create Session Request/ Response	MME→S-GW→P-GW	Establish PDN connection
Modify Bearer Request/ Response	MME→S-GW	Update after HO (new eNB TEID)
Delete Session Request/ Response	MME→S-GW→P-GW	Release PDN connection
Create Bearer Request/Response	P-GW→S-GW→MME	Dedicated bearer setup
Delete Bearer Request/Response	P-GW→S-GW→MME	Dedicated bearer release
Release Access Bearers	MME→S-GW	UE goes IDLE (release S1-U)
Downlink Data Notification	S-GW→MME	DL data arrived for idle UE

GTPv2-C Header:

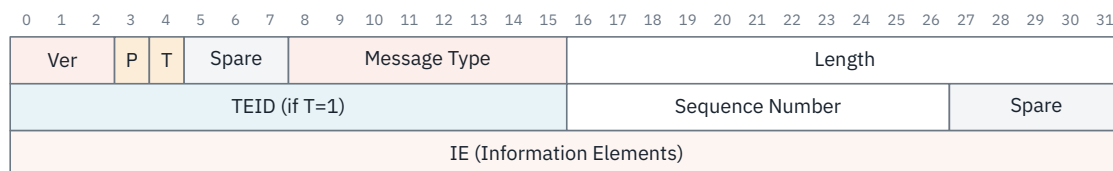


Figure 24.6 GTPv2-C header

6.3 5G: From GTP-C to PCF + HTTP/2

In 5G, the traditional GTP-C role is split:

- **SMF ↔ UPF:** Uses **PCF** (N4) — not GTP-C
- **Inter-NF signaling:** Uses **HTTP/2 SBI** — not GTP-C
- GTPv2-C remains for interworking with 4G (N26 interface: AMF↔MME)

7 HTTP/2 SBI (Service-Based Interface)

7.1 Concept

5G Core replaces point-to-point interfaces with a **Service-Based Architecture (SBA)**:

- Each NF exposes **services** via RESTful APIs
- Communication uses **HTTP/2** over **TLS 1.3** over **TCP**
- Data format: **JSON** (OpenAPI 3.0 specifications)

7.2 Key NF Services

NF	Service	Operations	Purpose
AMF	Namf_Communication	N1N2MessageTransfer, UEContextTransfer	NAS relay, context
SMF	Nsmf_PDUSession	Create, Update, Release	PDU session lifecycle
UDM	Nudm_SubscriberDataMgmt	Get, Subscribe, Unsubscribe	Subscriber data
AUSF	Nausf_UEAuthentication	Authenticate	5G-AKA, EAP-AKA'
PCF	Npcf_SMPolicyControl	Create, Update, Delete	QoS/charging policy
NRF	Nnrf_NFDiscovery	Discover	Find NF instances
NSSF	Nnssf_NSSelection	Get	Network slice selection
NEF	Nnef_EventExposure	Subscribe, Unsubscribe	Event reporting to AF

7.3 Communication Patterns

Pattern	Description	Example
Request-Response	Synchronous API call	SMF queries UDM for subscription
Subscribe-Notify	Asynchronous event	PCF subscribes to AMF for UE location
Service Discovery	Query NRF for NF endpoint	AMF discovers SMF via NRF

7.4 SBI Protocol Stack

NF Service (e.g., Nsmf)
HTTP/2 (multiplexed streams)
TLS 1.3 (mTLS between NFs)
TCP
IP

Figure 24.7 SBI protocol stack

8 Signaling Protocol Evolution: SS7 → Diameter → HTTP/2

8.1 Timeline

Era	Technology	Signaling Stack	Characteristics
1G/2G (1980s-90s)	GSM	SS7 (MAP, ISUP, SCCP, MTP)	Circuit-switched, TDM-based, binary encoding
3G (2000s)	UMTS	SS7 + RANAP + GTP-C v1	IP transport added (SIGTRAN), still SS7 app layer
4G (2010s)	LTE	Diameter + S1AP + GTP-C v2	All-IP, AVP-based encoding, point-to-point
5G (2020s)	NR	HTTP/2 SBI + NGAP + PFCP	Cloud-native, RESTful, service-mesh, JSON

8.2 Key Evolutionary Drivers

SS7 (1980s)	Diameter (2010s)	HTTP/2 SBI (2020s)
TDM transport	IP transport	IP transport
Binary encoding	AVP binary encoding	JSON text encoding
Fixed nodes	Fixed nodes	Microservices (NFs)
Point-to-point	Point-to-point	Service mesh
Centralised (STP)	Diameter agents (DRA)	Service discovery (NRF)
Rigid interfaces	Defined interfaces	API-driven, extensible
Hard to scale	Moderately scalable	Cloud-native, auto-scale
Telecom-only	Telecom-specific	IT/cloud standard

8.3 Signaling Evolution Timeline Diagram

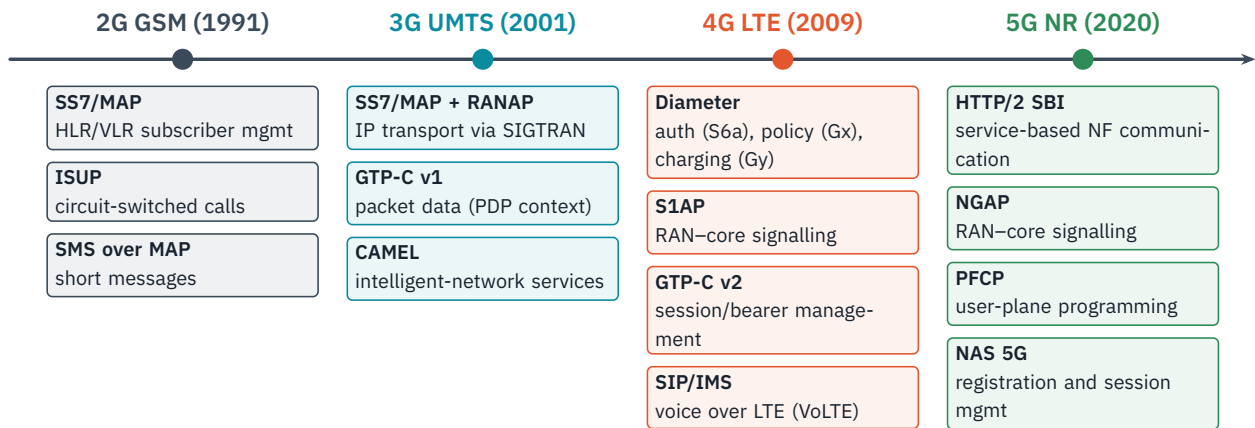


Figure 24.8 Evolution of mobile signalling protocols

8.4 Protocol Comparison: Auth Flow Example

2G/3G (SS7/MAP):

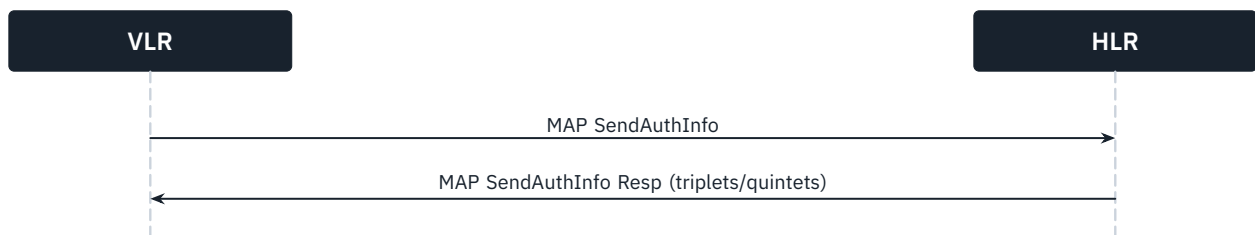


Figure 24.9 2G/3G authentication over SS7/MAP

4G (Diameter):

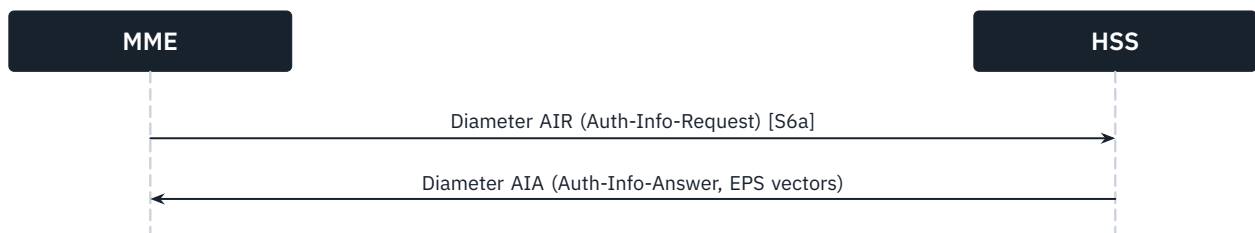


Figure 24.10 4G authentication over Diameter (S6a)

5G (HTTP/2):

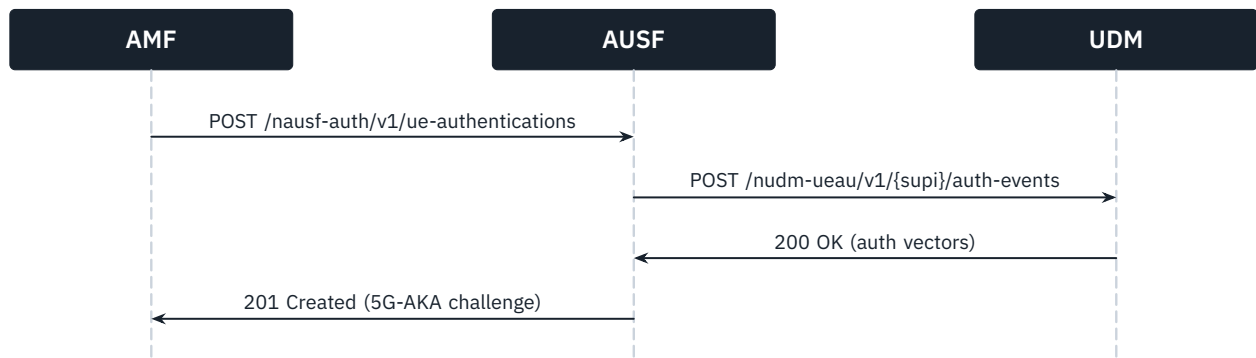


Figure 24.11 5G authentication over HTTP/2 SBI

Complete Control Plane Architecture Diagram (5G)

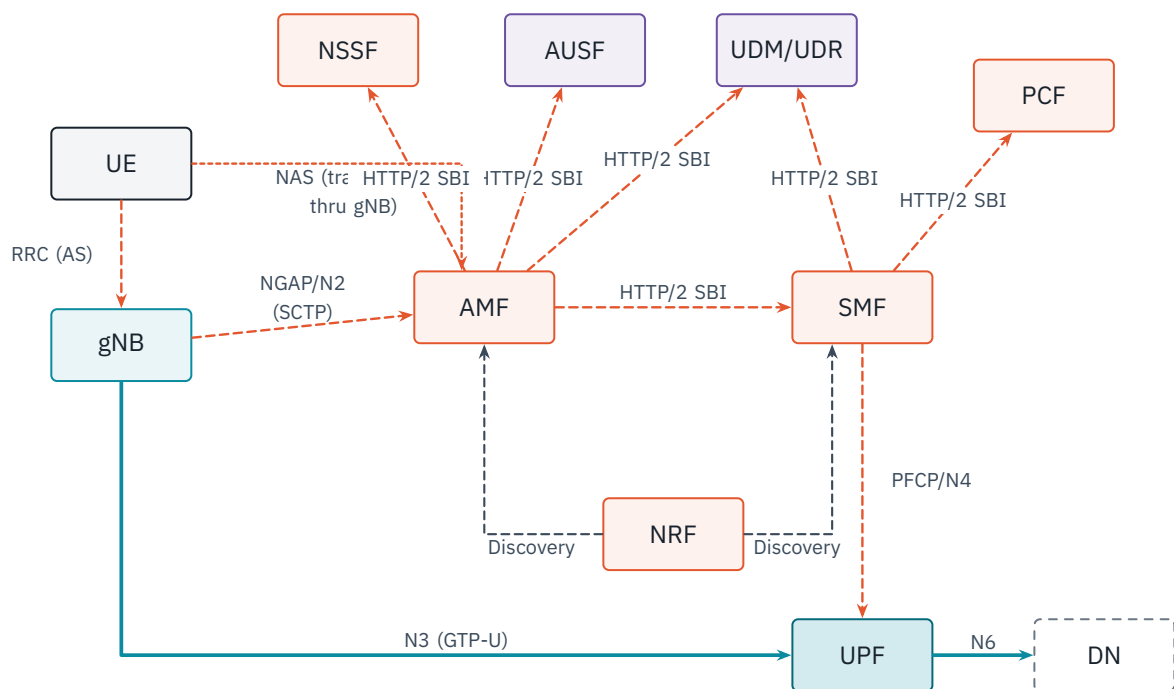


Figure 24.12 Complete 5G control-plane architecture

Summary: Control Plane vs. User Plane

Aspect	Control Plane	User Plane
Carries	Signaling messages	User data (IP, voice, video)
Purpose	Setup, manage, release connections	Transport data end-to-end
Latency tolerance	Moderate (100s of ms OK)	Critical (low latency required)
Volume	Low (small messages)	High (bulk data transfer)
Protocols	NAS, RRC, NGAP, HTTP/2, Diameter	GTP-U, SDAP, PDCP
5G separation	AMF, SMF, AUSF, etc.	UPF (controlled by SMF via PFCP)
Scaling	Scale per signaling load	Scale per throughput demand
State	Stateful (tracks UE context)	Stateless forwarding (rules from CP)

Key Takeaways

1. **Control plane = signaling brain:** Sets up tunnels, authenticates users, manages mobility
2. **NAS is end-to-end (UE↔Core)** — transparent to RAN; separate security context
3. **RRC manages the radio** — connection setup, measurements, handover commands
4. **5G adds RRC_INACTIVE** state for fast resume and IoT efficiency
5. **NGAP replaced S1AP** — supports slicing, QoS flows, and three RRC states
6. **GTP-C evolved:** v1 (3G) → v2 (4G) → replaced by PFCP+HTTP/2 in 5G
7. **HTTP/2 SBI is revolutionary** — cloud-native, RESTful, auto-scalable NF communication
8. **Signaling evolution:** SS7 (binary, TDM) → Diameter (binary, IP) → HTTP/2 (JSON, cloud-native)